

AGGREGATE TRENDS PRESENTATION 2

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Thursday, October 23rd



Redirection Analysis

Understanding how and why calls are redirected for the top called numbers.

1. Define Webex redirection fields
2. Example call path
3. Redirect vs. original reasons (top numbers)
4. Granular front desk analysis
5. Recommendations
6. Next Steps

Each call can generate multiple legs, redirections help explain how calls move through the system.

Call Start

- **Calling Number** → number that initiates the call
- **Direction = TERMINATING** → indicates the call is inbound
- **Correlation ID** → groups all call legs of the same session

Call Movement

- **Called number** → initial recipient
- **Redirecting Number** → who last redirected the call
- **Original Reason** → first trigger for a redirection
- **Redirect Reason** → why subsequent redirection occurs
- **Related Reason** → the trigger that led to a change in call presence

Call End

- **User Type** → identifies the endpoint (User, Voicemail, etc.)
- **Answered** → whether the call was answered or not
- **Report Time** → when the call ended

Call Path for Correlation ID '9108b891-9c8f-4056-a98c-dea3d0a121a8' (March 2025).

Client initially calls the Ombudsman Nursing Home line, which in turn tries to reach the staff directory (English option)

Called number =
13123478300, Staff directory

Direction =
Terminating

Original reason = No
Answer

Redirecting number =
13122296079, Nursing
home

Redirect reason = No
Answer

User type = Virtual line

Related reason = Call
Forward No Answer

Model = Cisco 8841

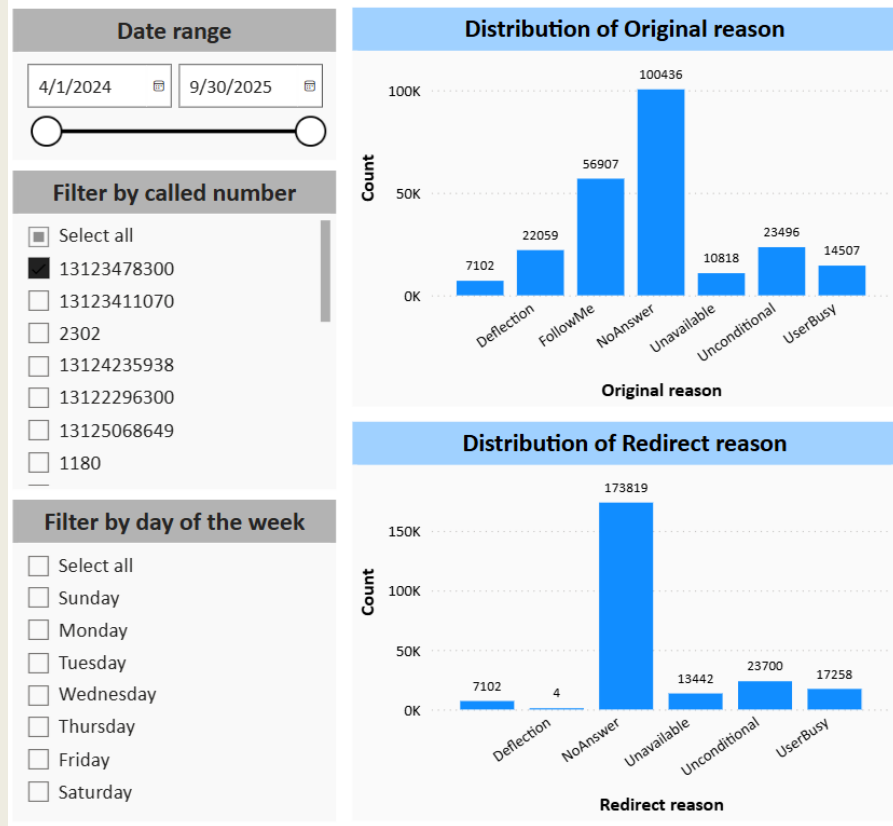
User = Voice Mail Retrieval

Call outcome = SUCCESS

Answered = TRUE

*The redirected call lands on **an Ombudsman intake line**, which is a real staff endpoint (not voicemail). The **Cisco 8841** model indicates a physical Webex desk phone used by the staff member who answered.*

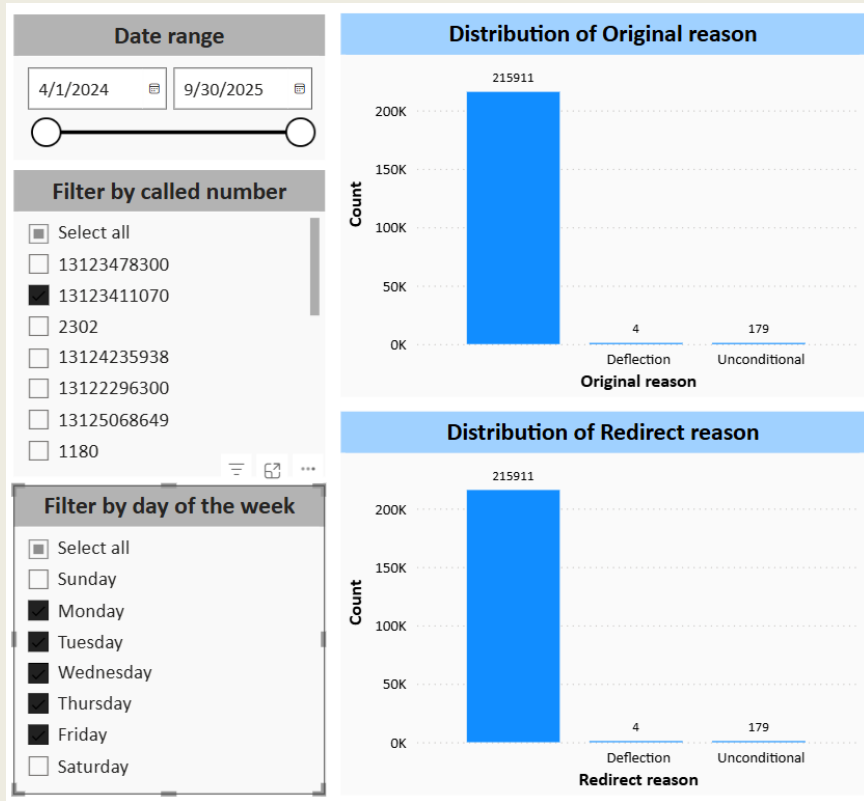
Distributions of original/redirect reasons for internal voicemail (13123478300) reflect its role as the ending leg of failed calls.



What We Found:

- Redirect reason (and original reason) is **predominantly NoAnswer** for internal voicemail, reflecting how it is the **outcome of failed calls**.
- Deflection** is an **extremely rare redirect reason** for 8300 → it is **not the intended outcome**
- A call ending in 8300 could be used as an **indication of failure** → possibly helpful in calculating success rate metrics

Most phone numbers do not significantly differ in their original/redirect distributions between weekday calls and weekend calls.



What We Found:

- For many of the phone numbers that I created as a filter, there is no major difference between their reason distributions on weekdays versus weekends
- Specifying hours of the day to look at will likely be more fruitful than trying to analyze the data at the level of days of the week

Date range

4/1/2024

9/30/2025

Filter by called number

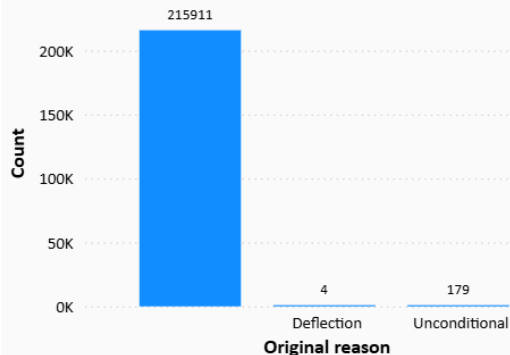
- ☒ Select all
- ☐ 13123478300
- ☒ 13123411070
- ☐ 2302
- ☐ 13124235938
- ☐ 13122296300
- ☐ 13125068649
- ☐ 1180



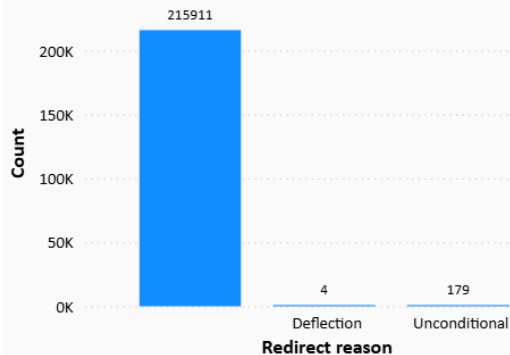
Filter by day of the week

- ☒ Select all
- ☐ Sunday
- ☒ Monday
- ☒ Tuesday
- ☒ Wednesday
- ☒ Thursday
- ☒ Friday
- ☐ Saturday

Distribution of Original reason



Distribution of Redirect reason



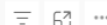
Date range

4/1/2024

9/30/2025

Filter by called number

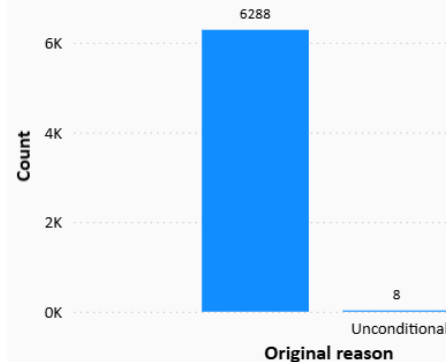
- ☒ Select all
- ☐ 13123478300
- ☒ 13123411070
- ☐ 2302
- ☐ 13124235938
- ☐ 13122296300
- ☐ 13125068649
- ☐ 1180



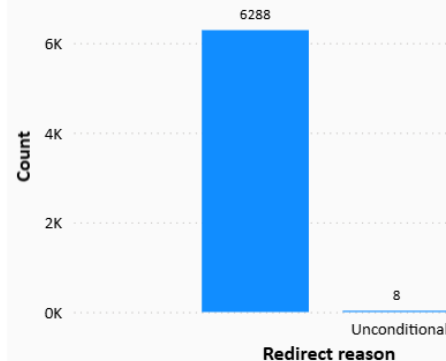
Filter by day of the week

- ☒ Select all
- ☒ Sunday
- ☐ Monday
- ☐ Tuesday
- ☐ Wednesday
- ☐ Thursday
- ☐ Friday
- ☒ Saturday

Distribution of Original reason

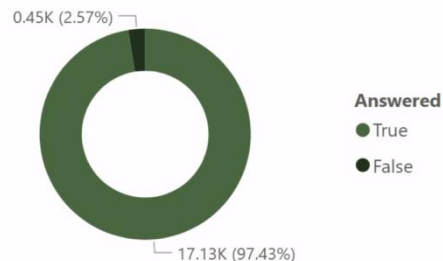


Distribution of Redirect reason

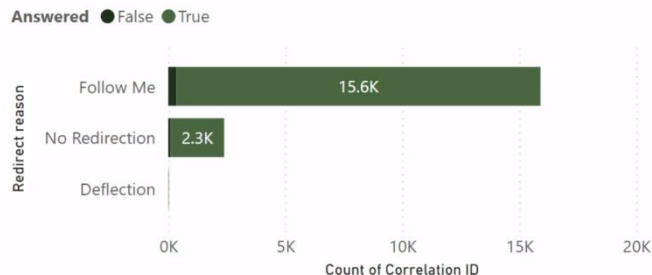


Front desk calls are being answered efficiently, with most actively managed through automatic routing.

Answered vs Not Answered (Front Desk)



Redirect reason for Front Desk Calls

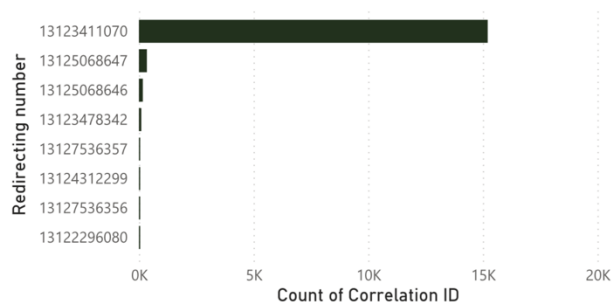


What We Found:

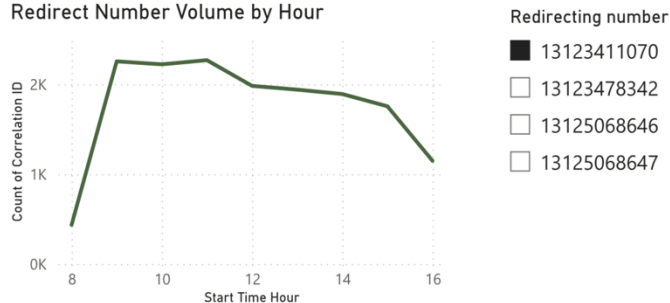
- ~ **97% of front desk calls are answered** showing strong coverage overall
- **The dominant redirect reason is “Follow Me”**, meaning the calls are automatically routed through Webex’s sequential forwarding logic until answered
- **Most redirected calls still reach a person**, showing that automated routing is working effectively to prevent missed calls for this line

Most “Follow Me” redirections from the front desk route through the main number, peaking mid-day and tapering after 4 PM.

Count of Correlation ID by Redirecting number



Redirect Number Volume by Hour



What We Found:

- 15.6k redirected calls route through the **main number (13123411070)**; it seems to act as the system hub
- Specialized menus handle smaller volumes, **Spanish (13125068647)** ~ 347 calls & **English (13125068646)** ~ 170 calls
- Redirect call volumes for the main line from the front desk peak between **9 AM and 2 PM**, aligning with business hours

Recommendations

1

Reallocate agent availability to areas that often result in “No Answer”. This would reduce the rate of calls ending in voicemail. However, practically, this might be infeasible due to costs.



2

Add bilingual auto-menu routing at the front-desk number to reduce unnecessary “Follow Me” transfers. A brief bilingual menu at the front desk entry point would let callers self-select their language immediately, decreasing redirections from this line.



Questions & Next Steps

Open Questions

- What is the Follow Me routing logic in Webex? Can the configuration be accessed directly or can we only infer it through All Call data?
- What are the use cases for callers dialing the front desk directly instead of the main line?

Next Steps

1. **Analyze Patterns Behind Voicemail.** Identify trends in **Reason**, **User Type** and **Client Type** for calls that end in extension 8300 (internal voicemail) to isolate whether missed calls are from staffing gaps or menu inefficiencies.
2. **Expand Use of “Related Reason” Field.** Investigate distinctions between **Deflection** and **Consultative Transfer** to better understand how and why front-desk calls move between lines.

Submenu Analysis

Hierarchical Menu Usage Analysis: Power BI Drill-Down Dashboard



Defining the Hierarchy and Variables Used

EP (Entry Point)

Main menu where a call begins (e.g. Legal Menu, Housing Menu).

Flow

A submenu or routing option under an EP.

Activity

The final action or step reached within a call (e.g. Language Selection, Queue).

Contact Session ID

Unique identifier for each distinct call. Counted each time the ID is within a menu.

Submenu Count

Number of submenus navigated during a single call.

Unknown Values

Placeholder for missing data. Ensures complete hierarchy display.

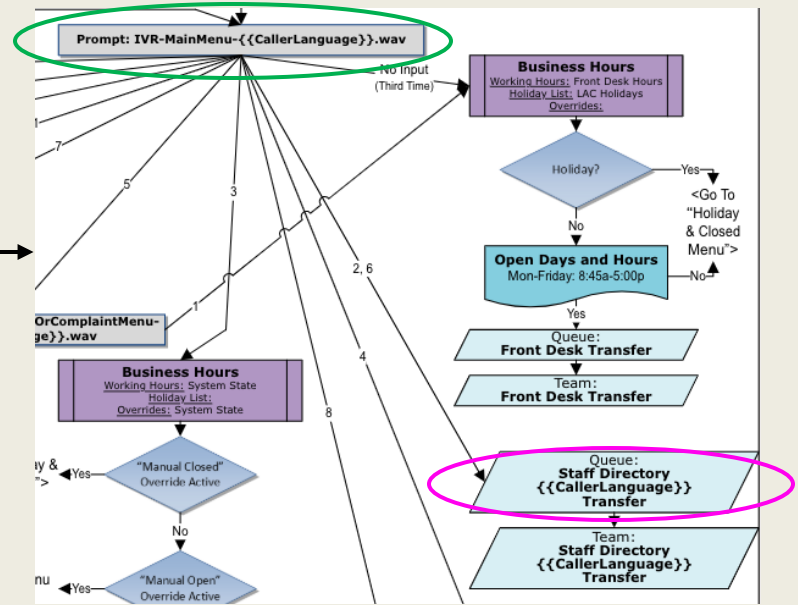
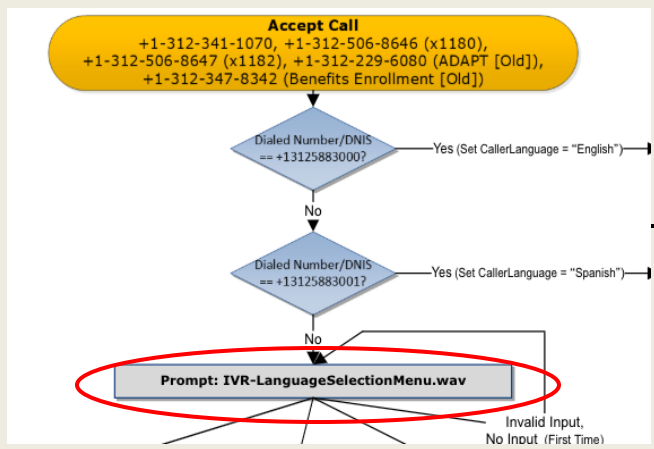
01	Contact Session ID	EP Name	Flow Name	Activity Name	Activity Start/End Time	
0001	0013748-8500-4550-8641-335479838600	Main Number Telephone EP			1/14/25 12:39	12
0001	0013748-8500-4550-8641-335479838600		LACMain		1/14/25 12:39	
0001	0013748-8500-4550-8641-335479838600	Main Number Telephone EP	LACMain	LanguageSelectMenu	1/14/25 12:39	12
0001	0013748-8500-4550-8641-335479838600	Main Number Telephone EP			1/14/25 12:40	12
0001	0013748-8500-4550-8641-335479838600	Main Number Telephone EP		MainMenu	1/14/25 12:39	12
0001	0013748-8500-4550-8641-335479838600			PreLegalMenuSeniorMenu	1/14/25 12:40	
0001	0013748-8500-4550-8641-335479838600	Pre-Legal Menu Senior Menu	LegalMenu	SeniorMenu	1/14/25 12:40	
0001	0013748-8500-4550-8641-335479838600	Legal Menu Telephone EP	LegalMenu		1/14/25 12:40	
0001	0013748-8500-4550-8641-335479838600	Legal Menu Telephone EP	LegalMenu	LegalMenu2	1/14/25 12:40	12
0001	0013748-8500-4550-8641-335479838600		LegalEmploymentMenu		1/14/25 12:43	
0001	0013748-8500-4550-8641-335479838600	Legal Employment Menu Telephone EP		EmploymentMenu	1/14/25 12:43	
0001	0013748-8500-4550-8641-335479838600			LegalOS-12-43		
0001	0013748-8500-4550-8641-335479838600		ClosedQueueMenu		1/14/25 12:43	12
0001	0013748-8500-4550-8641-335479838600	Closed Queue Menu Telephone EP		ClosedQueueMenu	1/14/25 12:43	12
0001	0013748-8500-4550-8641-335479838600	Closed Queue Menu Telephone EP			1/14/25 12:44	
0001	0013748-8500-4550-8641-335479838600			LegalOS-12-44		
0001	00223931-8614-4661-0160-0780760604	Main Number Telephone EP			1/14/25 14:50	14
0001	00223931-8614-4661-0160-0780760604		LACMain		1/14/25 14:50	14
0001	00223931-8614-4661-0160-0780760604	Main Number Telephone EP		LanguageSelectMenu	1/14/25 14:50	14
0001	00223931-8614-4661-0160-0780760604	Main Number Telephone EP	LACMain		1/14/25 14:50	14
0001	00223931-8614-4661-0160-0780760604	Main Number Telephone EP		MainMenu	1/14/25 14:50	14
0001	00223931-8614-4661-0160-0780760604			PreLegalMenuSeniorMenu	1/14/25 14:50	14
0001	00223931-8614-4661-0160-0780760604	Pre-Legal Menu Senior Menu	LegalMenu	SeniorMenu	1/14/25 14:50	14
0001	00223931-8614-4661-0160-0780760604	Legal Menu Telephone EP	LegalMenu		1/14/25 14:50	14
0001	00223931-8614-4661-0160-0780760604	Legal Menu Telephone EP	LegalMenu	LegalMenu2	1/14/25 14:51	14
0001	00223931-8614-4661-0160-0780760604		LegalFamilyMenu		1/14/25 14:52	14
0001	00223931-8614-4661-0160-0780760604	Legal Family Menu Telephone EP		FamilyMenu	1/14/25 14:52	14
0001	00223931-8614-4661-0160-0780760604			Queue	1/14/25 14:52	
0001	00223931-8614-4661-0160-0780760604		ClosedQueueMenu		1/14/25 14:52	14
0001	00223931-8614-4661-0160-0780760604	Closed Queue Menu Telephone EP		ClosedQueueMenu	1/14/25 14:52	14
0001	00223931-8614-4661-0160-0780760604	Closed Queue Menu Telephone EP			1/14/25 14:53	14
0001	00223931-8614-4661-0160-0780760604			ClosedQueueMenu	1/14/25 14:53	14
0001	00265673-0053-4274-0404-ee8bba67288b	Main Number Telephone EP			1/13/25 05:04	

#	contact_session_id	name	flow_name	activity_name	activity_start_timestamp	hour	submenu_count
1	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	Unknown Flow	Unknown Activity	2025-09-13T12-51:02	12	10
2	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	Unknown Activity	2025-09-13T12-51:02	12	10
3	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	LanguageSelection	2025-09-13T12-51:02	12	10
4	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	Unknown Activity	2025-09-13T12-51:02	12	10
5	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	MainMenu	2025-09-13T12-51:02	12	10
6	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	StaffDirectoryEnglis	2025-09-13T12-51:02	12	10
7	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	Unknown Activity	2025-09-13T12-51:02	12	10
8	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	Unknown Activity	2025-09-13T12-51:02	12	10
9	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	Unknown Activity	2025-09-13T12-51:02	12	10
10	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	Unknown Activity	2025-09-13T12-51:02	12	10
11	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	Unknown Activity	2025-09-13T12-51:02	12	10
12	00000242-f51f-458b-826e-cfa53bf69f	Main Number Telephone EP	LACMain	Unknown Activity	2025-09-13T12-52:02	12	10
13	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Main Number Telephone EP	Unknown Flow	Unknown Activity	2025-02-28T08-33:02	8	15
14	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Main Number Telephone EP	LACMain	Unknown Activity	2025-02-28T08-33:02	8	15
15	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Main Number Telephone EP	LACMain	LanguageSelection	2025-02-28T08-33:02	8	15
16	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Main Number Telephone EP	LACMain	Unknown Activity	2025-02-28T08-33:02	8	15
17	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Main Number Telephone EP	LACMain	PreLegalMenuSeniorsMenu	2025-02-28T08-33:02	8	15
18	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Pre-Legal Menu Seniors Menu Telephone / PreLegalMenuSeniorsMenu	SeniorMenu	SeniorMenu	2025-02-28T08-34:02	8	15
19	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Pre-Legal Menu Seniors Menu Telephone / LegalMenu	Unknown Activity	2025-02-28T08-34:02	8	15	
20	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Legal Menu Telephone EP	LegalMenu	LegalMenu	2025-02-28T08-34:02	8	15
21	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Legal Menu Telephone EP	LegalMenu	LegalMenu	2025-02-28T08-35:02	8	15
22	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Legal Menu Telephone EP	LegalFamilyMenu	Unknown Activity	2025-02-28T08-35:02	8	15
23	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Legal Family Menu Telephone EP	LegalFamilyMenu	FamilyMenu	2025-02-28T08-35:02	8	15
24	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Queues	Queues	Unknown Activity	2025-02-28T08-35:02	8	15
25	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Closed QueueMenu	ClosedQueueMenu	ClosedQueueMenu	2025-02-28T08-35:02	8	15
26	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Closed QueueMenu Telephone EP	ClosedQueueMenu	ClosedQueueMenu	2025-02-28T08-35:02	8	15
27	00014548-a6ce-acb2-a529-d55e2c9c3d4a	Closed QueueMenu Telephone EP	ClosedQueueMenu	Unknown Activity	2025-02-28T08-35:02	8	15
28	00023447-6772-a62f-b26f-85591f1085f6	Main Number Telephone EP	Unknown Flow	Unknown Activity	2024-06-29T10-55:02	10	5
29	00023447-6772-a62f-b26f-85591f1085f6	Main Number Telephone EP	LACMain	Unknown Activity	2024-06-29T10-55:02	10	5
30	00023447-6772-a62f-b26f-85591f1085f6	Main Number Telephone EP	LACMain	LanguageSelection	2024-06-29T10-55:02	10	5
31	00023447-6772-a62f-b26f-85591f1085f6	Main Number Telephone EP	LACMain	Unknown Activity	2024-06-29T10-55:02	10	5
32	00023447-6772-a62f-b26f-85591f1085f6	Main Number Telephone EP	LACMain	MainMenu	2024-06-29T10-55:02	10	5
33	00023447-6772-a62f-b26f-85591f1085f6	Main Number Telephone EP	LACMain	Unknown Activity	2024-06-29T10-56:02	10	5
34	00039636-5377-4488-95a-779af5dc62c2	Main Number Telephone EP	Unknown Flow	Unknown Activity	2024-09-25T10-49:02	10	5
35	00039636-5377-4488-95a-779af5dc62c2	Main Number Telephone EP	LACMain	Unknown Activity	2025-09-12T10-49:02	10	5

- Standardized/combined CAR dataset
- Created submenu_count variable
- Replaced NAs with “Unknown” placeholders
- Converted ep_name, flow_name, and activity_name to text
- Filled missing EPs and Flows by filling from the row above and grouping by contact session id

- Standardized/combined CAR dataset
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- Replaced NAs with “Unknown” placeholders
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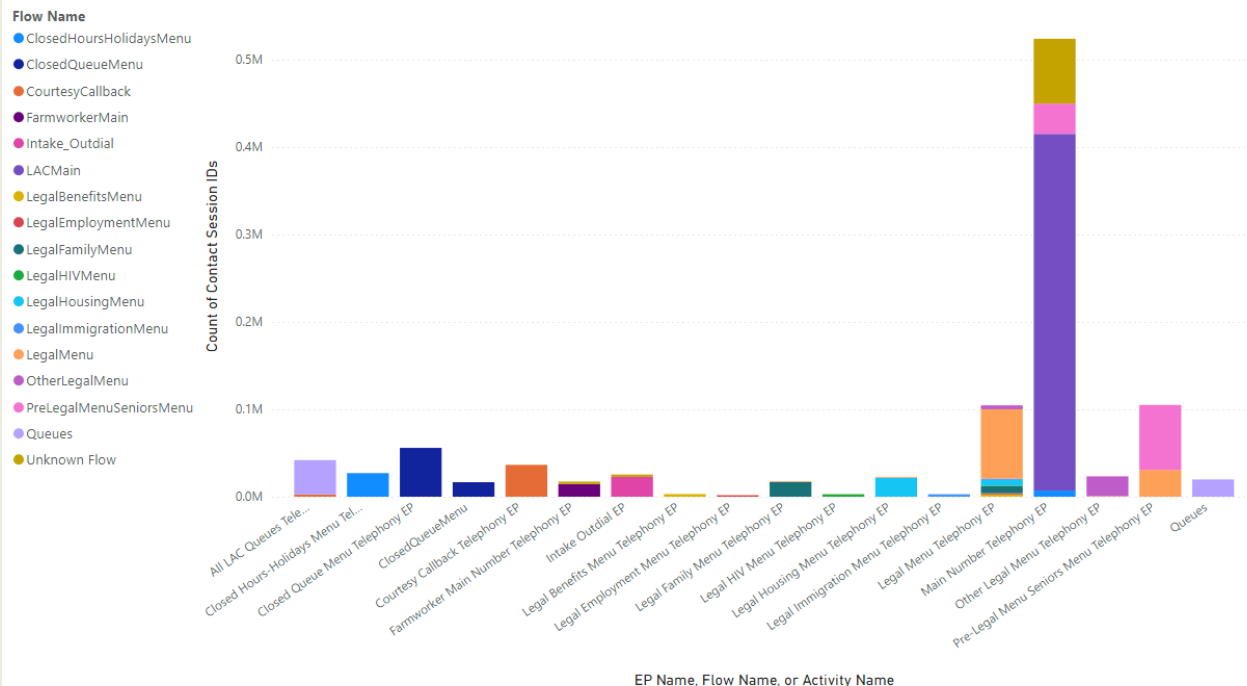
Call Flow Example



	contact_session_id	ep_name	flow_name	activity_name	activity_start_timestamp	hour	submenu_count
1	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	Unknown Flow	Unknown Activity	2025-03-13T12:51:00Z	12	10
2	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	Unknown Activity	2025-03-13T12:51:00Z	12	10
3	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	LanguageSelection	2025-03-13T12:51:00Z	12	10
4	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	Unknown Activity	2025-03-13T12:51:00Z	12	10
5	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	MainMenu	2025-03-13T12:51:00Z	12	10
6	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	StaffDirectoryEnglish	2025-03-13T12:51:00Z	12	10
7	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	StaffDirectoryEnglish	2025-03-13T12:51:00Z	12	10

Baseline Distribution of Contact Session IDs by EP Name, Colored by Flow Name

Count of Contact Session IDs by EP Name, Flow Name, and Activity Name



What We Found:

- The **Main EP** is visited most frequently; indicates we should prioritize simplifying this EP since it takes extra time to read out all these options
- Under the Main Menu, there are **12 Activity Names**, which takes **58 seconds** to list; separating these activities into 2 groups could improve efficiency
- Benefits, Employment, HIV, Immigration EPs** are rarely used; could combine these to simplify the system then have them be just flows instead of their own EPs

Recommendations

1

Simplify high-complexity menus



2

Develop Menu Efficiency Metrics



3

Reduce Queue-Related Bottlenecks



Next Steps & Further Questions

- Add interactive filters and KPIs: Include slicers for date, day type, and hours; add cards for total calls and average submenus per EP
 - Develop efficiency indicators: Calculate average submenu depth per EP or Flow to highlight areas with the most caller confusion or unnecessary steps
 - Group related menus: Merge similar Eps (e.g. Benefits, Employment, HIV, Immigration) for clearer visualization and comparison
 - Group related activities: there are 12 options inside of Main Menu → splitting into two groups could improve efficiency
1. Does the EP → Flow → Activity hierarchy in our cleaned data match LAC's structure?
 2. Can Cynthia clarify what the Flow Name means?
 3. Would LAC be open to restructuring or merging high-friction menus based on our findings?
 4. Which metrics are most important to LAC between minimizing submenu steps, reducing queue time, or ensuring correct routing on the first try?

Appendix

** Includes additional details and visuals from our analysis that were not included in the live presentation due to time constraints.*



Data Cleaning (Python)

The following data cleaning was completed in **Python** for the **All Calls Dataset**. The new data from *September 2025* was also added to the combined code csv. Code [here](#).

Converting date to datetime format

```
In [59]: df_main["Start time"] = pd.to_datetime(df_main["Start time"], utc=True)
```

```
In [60]: df_main.to_csv("combined_calls.csv", index=False)
```

```
In [61]: df_main["Start time"] = df_main["Start time"].dt.tz_convert("America/Chicago").dt.tz_localize(None)
```

```
In [62]: df_main["Start time"].head()
```

```
Out[62]: 0    2024-04-30 18:58:53.988
1    2024-04-30 18:56:37.386
2    2024-04-30 18:54:59.099
3    2024-04-30 18:54:59.099
4    2024-04-30 18:54:52.336
Name: Start time, dtype: datetime64[ns]
```

```
In [63]: # Extract hour of day
df_main["Start Time Hour"] = df_main["Start time"].dt.hour
print(df_main[["Start time", "Start Time Hour"]].head())
```

	Start time	Start Time Hour
0	2024-04-30 18:58:53.988	18
1	2024-04-30 18:56:37.386	18
2	2024-04-30 18:54:59.099	18
3	2024-04-30 18:54:59.099	18
4	2024-04-30 18:54:52.336	18

```
In [64]: df_main.to_csv("combined_calls_with_start_time_hour.csv", index=False)
```